

Lemma G — the arc register of the spatial tilt

PLANETVORTEX — the point monograph

Table of Contents

Statement

Let i_i be the J2000 inclination of the body i (the committed JPL register) and $\dot{R}=1$ AU the register radius. Then $\lambda_i=\dot{R}i_i[r a d]$ is an exact, monotone, dimension-free measure of the spatial tilt, and the tilt operator $T_i=R_z(\Omega_i)R_x(i_i)\in SO(3)$ sends the ecliptic normal to the orbit normal $n_i=(\sin i_i \sin \Omega_i, -\sin i_i \cos \Omega_i, \cos i_i)$ exactly.

Proof

T_i is a product of rotations, hence lies in $SO(3)$ — orthogonality and $\det=1$ are closed under multiplication. The image of $(0,0,1)$ under $R_z(\Omega)R_x(i)$:

$R_x(i)(0,0,1)=(0,-\sin i,\cos i)$; $R_z(\Omega)$ rotates the x y -components:

$(\sin i \sin \Omega, -\sin i \cos \Omega, \cos i)$. $\lambda=\dot{R}i$ is linear in the angle with a fixed positive constant — monotone by inspection: it is the arc length cut by the tilt angle on the register circle. $\square$

Computational certificate

Stage V1: the orthogonality at 2.2e-16, the normal recovery at 0.0, the monotone arc ladder, and the 3D J2000 run (energy 2.1e-13, the vector L at 4.7e-15, Kepler III at 1.0e-4 against the secular mean elements).